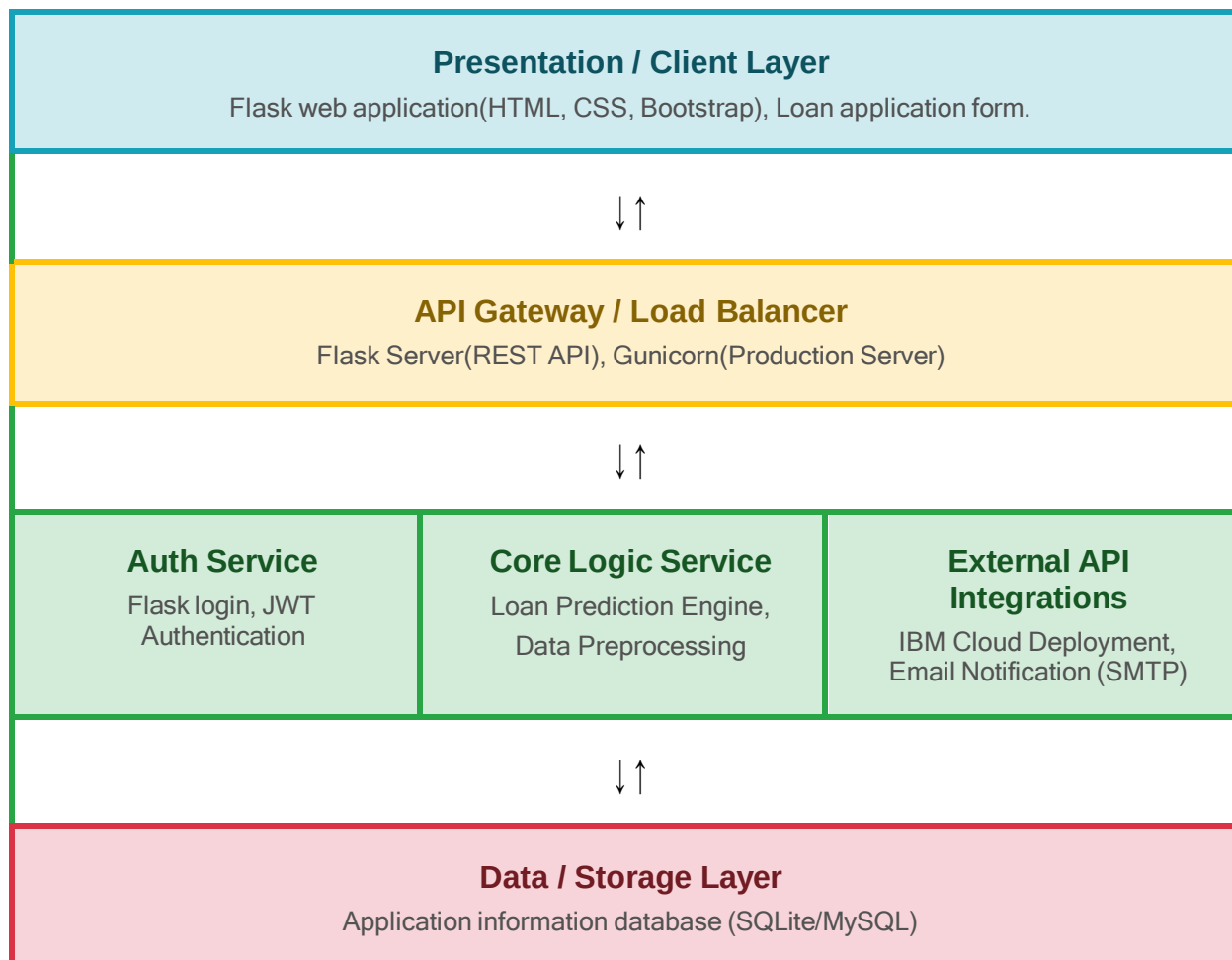


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	SmartLender-ApplicantCredibility Prediction for Loan Approval
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a web interface where bank officers or applicants enter loan details and receive instant loan approval predictions.	Flask, HTML, CSS, Bootstrap
API Gateway	Receives user requests, validates input, forwards data to the prediction service, and returns prediction results.	Flask REST API, Unicorn, Nginx
Microservices	Processes applicant data, performs preprocessing, loads the trained XGBoost model, predicts loan approval, and returns the result.	Python, Scikity-learn, XGBoost, Pandas, NumPy.
Database	Stores applicant details, datasets, trained ML model, and prediction history for future analysis.	SQLite/MySQL, Pickle(.pkl), CSV